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Department of Computer Science
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F.Y.B.Sc.(Data Science) Sem – II
DS-151-T : Advanced Python Programming

Question Bank

Unit1: File Handling

1. Which methods are used to read from a file? Explain any two with example
2. Write a program to read an entire text file.
3. Differentiate between:
   a. text file and binary file.
   b. readline() and readlines()
   c. write() and writelines()

4. Write the use and syntax for the following methods:
   a. open
   b. read()
   c. seek()
   d. tell()

5. Write the file mode that will be used for opening the following files. Also, write the Python statements to open the following files:
   a. text file "example.txt" in both read and write mode
   b. binary file "bfile.dat" in write mode
   c. text file "try.txt" in append and read mode
   d. binary file "btry.dat" in read only mode.

6. What is the difference between the following set of statements (a) and (b):
   a. P = open("practice.txt","r")
      P.read(10)

   b. with open("practice.txt", "r") as P:
      x = P.read()

7. Write a command(s) to write the following lines to the text file named hello.txt. Assume that the file is opened in append mode.
   a. "Welcome my class"
      "It is a fun place"
      "You will learn and play"

8. Write a command(s) to perform following on hello.txt file:
   a. To read data from 5th position in file.
   b. To insert data at a 10th specific Position in file.

c. To return current position of file pointer.

9. Write a Python program to open the file hello.txt used in question no 7 in read mode to display its contents. What will be the difference if the file was opened in write mode instead of append mode?

10. Write a Python Program to define following functions on "Demo.txt" file:

a. **Write_in_file() :** To write contents into file
b. **Read_from_file():** To read contents from file

Unit2: Python Libraries

1. List the visualization libraries used in python.
2. List the Statistical Analysis libraries used in python.
3. List the Deep Learning libraries used in python.
4. What is NumPy? Or What is the purpose of NumPy in Python?
5. Write a NumPy code snippet to create an array of zeros.
6. Write a NumPy code snippet to create an array of ones.
7. How do I create a NumPy array.
8. What are the main features of Numpy
9. How do I access elements in a NumPy array?
10. How do you reshape a NumPy array?
11. Explain Numpy attributes.
12. What is the shape attribute in a NumPy array?
13. How can you create a NumPy array from a Python list?
14. How to perform element-wise operations on NumPy arrays?
15. Define the min and max function in NumPy.
16. How can you access elements in a NumPy array based on specific conditions?
17. What is the difference between slicing and indexing in NumPy?
18. How do you sort a NumPy array in ascending or descending order?
19. What is difference between NumPy and Pandas?
20. What are ways of creating 1D, 2D and 3D arrays in NumPy
21. How do you find the mean, median, and standard deviation of a NumPy array.
22. How do you initialize an empty array or an array of zeros/ones in NumPy?
23. How do you initialize an empty array or an array of zeros/ones in NumPy?
24. Explain the difference between one-dimensional, two-dimensional, and multi-dimensional arrays.
25. Compare NumPy, Pandas or SciPy.

Unit3: Exception Handling

1. What is the use of pass statement?
2. Write any two common exceptions in Python.
3. Which are built in exceptions in Python?
4. What is the use of finally block?
5. List the syntax for handling exception.
6. Explain various types of exceptional handling in Python.
7. Explain Exception handling in python with example. (try-except-else) OR How to handle exception in Python?

8. Write the syntax of the raise statement & explain it. (Syntax, purpose, explanation, example). OR How to raise an exception with arguments.
9. Explain EXCEPT Clause with no exception.
10. Write a note on Custom Exception. (What Custom exception, Syntax, example)
11. Write a note on assert statement. (Syntax, purpose, explanation, example)
12. Write a Python program to check for Zero Division Error Exception.
13. Write a function that asserts the given number is positive.
14. Write a function that raises this exception if a negative number is passed.
15. Write a program to raise a user defined exception to check if age is less than 18.

Unit4: GUI Programming

1. Write the uses of Tkinter
2. Write the Steps to setup a Tkinter Application
3. List names of Tkinter Widgets.
4. Write a note on Geometry Management. Or Explain types of Geometry Management or Explain following Geometry Management types:
    a. Grid
    b. Place
    c. Pack
5. Explain Tkinter frame.
6. Explain Tkinter Button.
7. Explain Tkinter Label.
8. Explain Tkinter Entry.